Point Kinetics with Feedback

NE 630 | Nuclear Reactor Theory

29
LESSON

Read: FNRP 5.6 and 9.5

After a positive reactivity insertion, what determines whether neutron density settles at a higher level or rises to a peak and falls?

1 Start from critical equilibrium

For one precursor group and no external neutron source, the neutron density n and effective precursor concentration C obey

$$\dot{n} = \frac{\rho - \beta}{\Lambda} n + \lambda C, \qquad \dot{C} = \frac{\beta}{\Lambda} n - \lambda C.$$

At $t = 0^-$, $\rho = 0$, $n = n_0$, and $C = C_0 = \beta n_0/(\lambda\Lambda)$. A step insertion changes $\dot{n}$ at once. Both n and C are continuous.

Recall from Lesson 28. For constant ρ, try $n, C \propto e^{\omega t}$. Rearranging the characteristic equation gives the one-group inhour relation

$$\omega^2 + \left(\lambda - \frac{\rho - \beta}{\Lambda}\right)\omega - \frac{\lambda\rho}{\Lambda} = 0 \quad \Longrightarrow \quad \rho = \omega\left(\Lambda + \frac{\beta}{\omega + \lambda}\right).$$

When fuel heats up, ρ changes during the insertion. The instantaneous growth rate $\dot{n}/n$ then depends on both $\rho(t)$ and the precursor inventory.

2 Let the fuel temperature set reactivity

Write $\theta = T_f - T_{f0}$ for the fuel temperature rise and ρ_{ext} for the imposed reactivity. With $\theta(0) = 0$,

$$\boxed{\rho(t) = \rho_{\text{ext}}(t) + \alpha_f\,\theta(t)}, \qquad \alpha_f = -3 \times 10^{-5}\ \text{K}^{-1}.$$

A 100 K rise thus offsets a 300 pcm insertion. Here $1\ \text{pcm} = 10^{-5}$ in reactivity, and one dollar is β.

Write $p = n/n_0 = P/P_0$ for relative power at fixed flux shape and spectrum. A lumped heat balance with fixed coolant temperature T_c reads

$$C_{\text{th}}\dot{T}_f = P_0 p - H(T_f - T_c).$$

C_{th} is the fuel heat capacity (J/K) and H the fuel-to-coolant thermal conductance (W/K).

Model A. Fast heat removal. In the quasistatic limit, heat removed balances power at each instant. Subtract the initial balance $H(T_{f0} - T_c) = P_0$ to get

$$\theta = A(p - 1), \qquad \dot{\theta} = A\dot{p}, \qquad A = P_0/H.$$

Model B. Adiabatic fuel. Neglect heat removal during the transient. Deposited energy accumulates as sensible heat.

$$\dot{\theta} = Bp, \qquad \theta(t) = B\int_0^t p(s)\,ds, \qquad B = P_0/C_{\text{th}}.$$

Both cases start from the same critical neutron state. For B, assume cooling balanced P_0 before $t = 0$ and ceases at the start of the run. Thus $\dot{\theta}(0^+) = B$ even at $p = 1$.

3 Add one state to Lesson 28's solver

Use $\mathbf{y} = (n, C, \theta)$. The notebook supplies β, Λ, λ, n_0, and α_f. Set $A = 100$ K for the reference fuel-to-coolant temperature difference. Use $B = 240$ K/s, corresponding to $P_0/V \simeq 720$ W/cm^3 and $C_{\text{th}}/V \simeq 3$ J/(cm^3 K). The `fixed` case holds $T_f = T_{f0}$.

```
import numpy as np
from scipy.integrate import solve_ivp

beta, Lambda = 0.0065, 1e-4            # Lambda in s
lam = np.log(2) / 5.0                  # 1/s
alpha = -3e-5                          # 1/K
A, B = 100.0, 240.0                    # K, K/s
n0 = 1e9                               # 1/cm^3
y0 = [n0, beta*n0/(lam*Lambda), 0.0]  # n, C, theta

def f(t, y, rho_ext, model):
    n, C, theta = y
    rho = rho_ext + alpha*theta
    dn = (rho - beta)*n/Lambda + lam*C
    dC = beta*n/Lambda - lam*C
    if model == "A":
        dT = A*dn/n0
    elif model == "B":
        dT = B*n/n0
    elif model == "fixed":
        dT = 0.0
    else:
        raise ValueError("Use A, B, or fixed")
    return dn, dC, dT

def solve(rho_ext, times, model, start=y0):
    sol = solve_ivp(f, (times[0], times[-1]), start,
                    args=(rho_ext, model), t_eval=times,
                    method="LSODA", rtol=1e-8, atol=1e-9)
    if not sol.success:
        raise RuntimeError(sol.message)
    return sol.y
```

Apply the 300 pcm insertion for 5 s. Restart at the withdrawal time with the final state of the first solve. The stored fuel heat and precursor inventory carry over.

```
t1 = np.linspace(0, 5, 1001)
t2 = np.linspace(5, 10, 1001)
one = solve(0.003, t1, "B")
two = solve(0.0, t2, "B", one[:, -1])
```

Start a fresh prompt excursion. Hold $\rho_{\text{ext}} = 1.5\beta$ for the full interval shown. Replace `"B"` with `"A"` or `"fixed"` to compare the three curves on page 2.

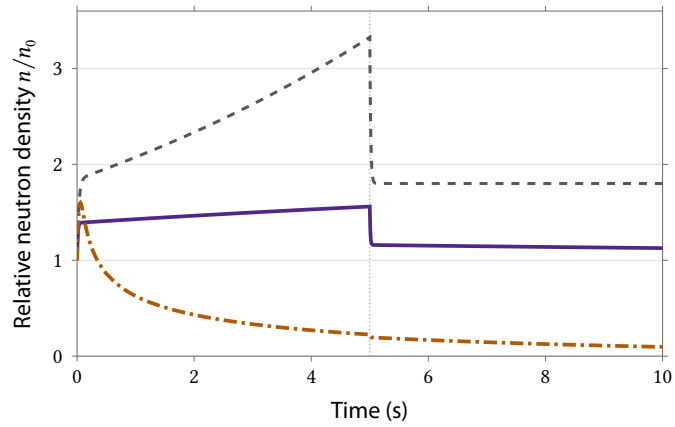
```
tp = np.linspace(0, 0.20, 501)        # seconds
n, C, theta = solve(1.5*beta, tp, "B")
p = n/n0
dollars = (1.5*beta + alpha*theta)/beta
```

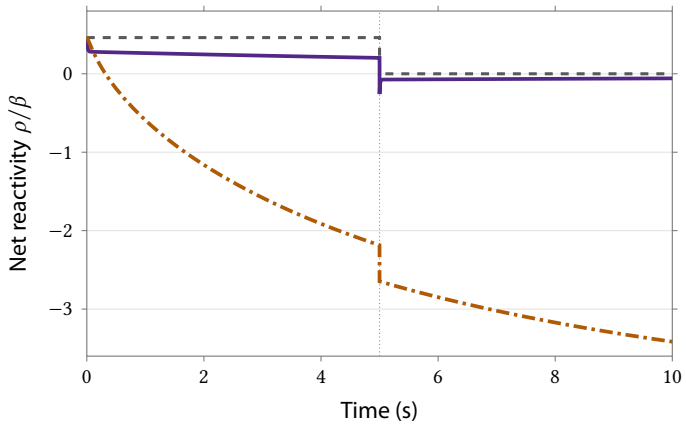
Two insertions, three temperature histories

Fixed T_f — Model A (fast heat removal) — Model B (adiabatic fuel)

4 Insert 300 pcm, then withdraw it at 5 s

The same positive insertion brings all three cases to $\rho/\beta = 0.462$ at $t = 0^+$. Read each density history alongside its net reactivity. The vertical guide marks withdrawal.

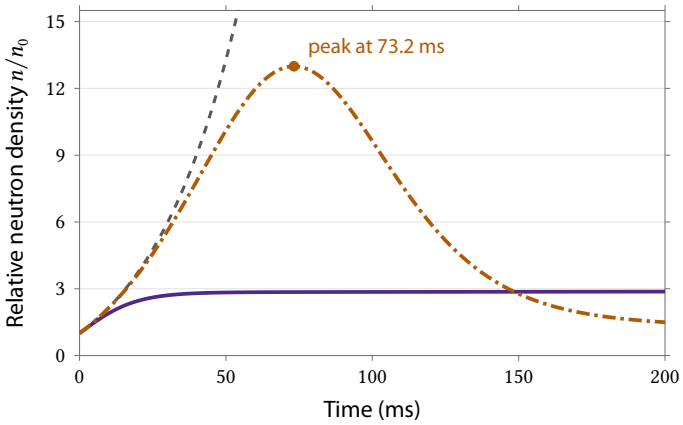


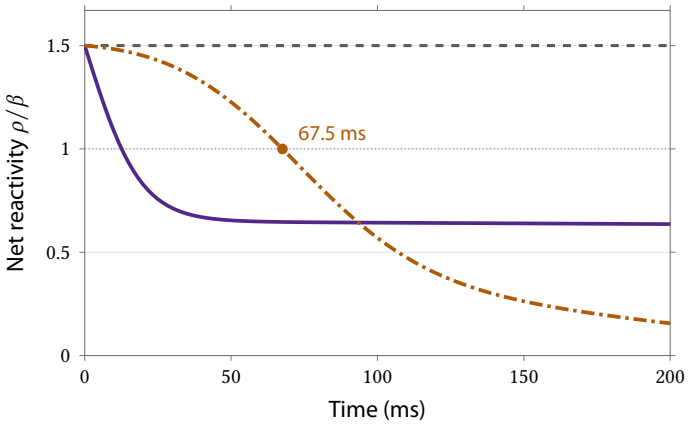


At withdrawal. Just before 5 s, n/n_0 is 3.33 at fixed T_f, 1.56 in A, and 0.225 in B. Removing the insertion lowers ρ by the same 300 pcm in every case. In Model A the fuel cools as power falls. In Model B it keeps heating as long as fissions deposit energy.

5 Hold a prompt-supercritical insertion of 1.50 dollars

Here $\rho_{\text{ext}} = 975$ pcm exceeds $\beta = 650$ pcm. At fixed temperature the prompt term has an e-folding time of $\Lambda/(\rho_{\text{ext}} - \beta) = 31$ ms. Follow the adiabatic curve from the prompt-critical crossing (right) to the density peak (left).





Why does the peak come later? At $\rho = \beta$ the prompt term vanishes, but $\dot{n} = \lambda C > 0$. Model B reaches this threshold after a fuel temperature rise of $(975 - 650)/3 \simeq 108$ K. Density peaks at $13.0\,n_0$ when $\rho/\beta \simeq 0.918$ and the prompt loss just balances λC. With T_f fixed, density has already reached $76.3\,n_0$ by 100 ms, above the plotted range.

Two checks on the curves

After withdrawal at fixed temperature, add the neutron and precursor equations to show that $\frac{d}{dt}(n + C) = \rho n/\Lambda$. Which inventory is conserved after 5 s? Use the inventory at withdrawal and $C_\infty = \beta n_\infty/(\lambda\Lambda)$ to explain why n settles near $1.80\,n_0$.

If the 1.50-dollar insertion stays in, a positive steady power requires $\rho = 0$. Use Model A to predict the eventual n/n_0. Which term in Model B prevents a steady fuel temperature at positive power?